Catalogue of Markov model specifications

Extracted from a doctoral thesis on stochastic branching and release processes

Neutral-terminology copy for automated review

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Every entry gives the thesis location, state space, transition rates, parameters and attached conventions.
Rates are transcribed verbatim from the chapter sources.

Contents

Scope and conventions

This document lists every stochastic model in the source thesis whose law is *specified*: a state space plus a transition mechanism (transition matrix, generator, jump rates, or transition kernel). Deterministic population equations are included only where the thesis derives them as exact projections of a stochastic model, and the entry says so. Derived results are quoted only where they are part of the model's working apparatus (characteristic roots, regime boundaries, working parameter points).

Time is continuous and chains are time homogeneous unless an entry says otherwise; $o(\Delta t)$ terms are suppressed. Per-capita rates multiply the relevant count to give total rates. Model numbers M1–M31 are used for cross-reference. Sections are cited by thesis chapter and section number only.

Chapter 1

Catalogue at a glance

	Model	State space	Where
M1	Simple random walk	$\mathbb{Z}$ (discrete time)	§2.2.1.1
M2	Galton–Watson process (general; binary case)	$\{0, 1, 2, \dots\}$ (discrete time)	§2.2.1.2; Ch. 3
M3	Poisson process	$\{0, 1, 2, \dots\}$	§2.2.2.2
M4	Yule process	$\{1, 2, \dots\}$	§2.2.2.3
M5	Linear birth–death process (incl. time-inhomogeneous rates)	$\{0, 1, 2, \dots\}$	§2.2.2.4; §2.2.3
M6	Birth–death–catastrophe process	$\{0, 1, 2, \dots\} \cup \{H\}$	§2.2.2.5; §4.2; §5.2.2
M7	Seasonal two-population process	$\{0, 1, \dots\}^2$	§2.2.3.1
M8	Logistic-environment type-count process	$\{1, 2, \dots\}$, driven by an ODE	§2.2.3.2; §9.2
M9	Gated release (coupled ODE–CTMC)	$[0, \infty)^2 \times \{0, 1\}$	§2.2.4.1
M10	Surging gated release	$[0, \infty)^2 \times \{0, 1\}$	§2.2.4.2
M11	Absorption-only model	$\{0, \dots, N\}$ pairs	§2.B.1
M12	Absorption–death model	compartment pairs	§2.3.3.2
M13	Absorption–birth–death model	compartment pairs	§2.B.2
M14	Chained immediate transfer ($\mu = 0$ birth–catastrophe chain)	one load per compartment, M compartments	§5.10; §8.2.1
M15	Budding comparator compartment	two independent event streams	§5.11; §6.2.2
M16	Standard two-variable population model (deterministic projection)	$(\mathcal{I}, \mathcal{V})$ ODEs	§1.5; §6.2.1
M17	Kernel-driven renewal system	age-structured means	§6.3–6.4
M18	Branching approximation of the early phase; flooding parameter L	offspring counts	§6.6
M19	Reset process (recurrent release)	$\{0, 1, 2, \dots\}$ + compartment removal	§6.7.1
M20	Partial release (boolean thinning)	mean-field stock system	§6.7.2
M21	Budding-kernel extension with sigmoid incidence	kernel pair + incidence	§6.7.3
M22	Two-type birth–death–conversion with type-specific catastrophe	$\mathbb{Z}^2_{\geq 0} \cup \{(\mathsf{R}, \mathsf{R})\}$	§7.2
M23	Replicator–suppressor process	$\{0, \dots\} \times \{0, \dots, N\}$	§8.3.2
M24	Fixed-removal extension of the chained process	release sizes + budget	§8.4
M25	Competitive-reversal chain (evolving resistance)	$\{0, \dots, N\}$ + resistances	§9.3.2
M26	Fitness-coupled type-origination variant	populations + fitness + type counts	§9.3.6

	Model	State space	Where
M27	Rise–run–ruin–rejuvenation (RRRR)	count + potential $(X_t, V(t))$	§9.4.2
M28	Multiplicative-dissipation automaton	$L \times L$ lattice + potentials	§9.5
M29	Birth–conversion model (incl. vitality variant)	$\{0, 1, \ldots\}^2$	§9.A
M30	Public-good accumulation process	count + good concentration	§9.B
M31	Three-environment birth–death ordering	three birth–death processes	§10.3.1

Chapter 2

Foundational processes (thesis Chapter 2)

2.1 M1: Simple random walk

Specified in §2.2.1.1.

State space. $\mathbb{Z}$, discrete time.

Transitions. Fix $q \in (0, 1)$:

$$P_{i,i+1} = q, \qquad P_{i,i-1} = 1 - q. \tag{2.1}$$

Moments and law. $\mathbb{E}(X_n) = X_0 + n(2q - 1)$, $\mathrm{Var}(X_n) = 4nq(1 - q)$; for deterministic X_0, $\mathbb{P}(X_n = X_0 + 2r - n) = \binom{n}{r} q^r (1 - q)^{n-r}$, $r = 0, 1, \ldots, n$.

Anchors. The embedded jump chain of M5 has $q = \lambda/(\lambda + \mu)$. With absorbing barriers 0 and M, the hitting probability of M before 0 is $h_i = (1 - \theta^i)/(1 - \theta^M)$ with $\theta = (1 - q)/q$, reducing to i/M at $q = \frac{1}{2}$.

2.2 M2: Galton–Watson branching process (general and binary)

Specified in §2.2.1.2; developed throughout thesis Chapter 3.

State space. $\{0, 1, 2, \ldots\}$, discrete time.

Transitions. With offspring law $\mathbb{P}(L = k) = p_k$ and mean $m = \mathbb{E}(L)$, from $Z_0 = 1$:

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \tag{2.2}$$

the L_i independent copies of L. Under the probability generating function $\phi(z) = \mathbb{E}\Big(z^L\Big)$ the generational PGFs compose: $\phi_{n+1} = \phi_n \circ \phi$, $\phi_0(z) = z$.

Binary case used throughout the thesis.

$$p_2 = p, \qquad p_0 = 1 - p, \qquad p_k = 0 \quad (k \notin \{0, 2\}), \tag{2.3}$$

with $m = 2p$, $\mathbb{E}(Z_n) = (2p)^n$, $\phi(z) = pz^2 + (1 - p)$, and the survival recursion

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1, \tag{2.4}$$

where $S_n = \mathbb{P}(Z_n > 0)$. For $0 < p \le \frac{1}{2}$ survival decays to zero; for $p > \frac{1}{2}$ it decreases to the fixed point $(2p - 1)/p$.

Chapter 3 apparatus. For $0 < p < \frac{1}{2}$ the constant $A(p)$ is defined as the limit of $S_n/(2p)^n$ (existence via a convergent infinite product; degenerate endpoint $A(0) = \frac{1}{2}$); the population conditioned on survival settles to a quasi-stationary law with mean $1/A(p)$; near criticality $A(p) \sim 2(1 - 2p)$; rescaling the survival recursion to the logistic map identifies $A(p)$ with a value of the associated Koenigs linearising function.

2.3 M3: Poisson process

Specified in §2.2.2.2.

State space. $\{0, 1, 2, \ldots\}$; sole non-zero transition rate

$$q_{n,n+1} = \alpha, \qquad n \ge 0. \tag{2.5}$$

Law. From $N_0 = 0$: $\mathbb{P}(N_t = n) = \mathrm{e}^{-\alpha t}(\alpha t)^n/n!$, with $\mathbb{E}(N_t) = \mathrm{Var}(N_t) = \alpha t$. Thinning with retention probability ϖ gives a Poisson process of rate $\varpi\alpha$.

2.4 M4: Yule process

Specified in §2.2.2.3.

State space. $\{1, 2, \ldots\}$ from a single founder; sole non-zero transition rate

$$q_{n,n+1} = \lambda n, \qquad n \ge 1. \tag{2.6}$$

Law. $\mathbb{P}(Y_t = n) = \mathrm{e}^{-\lambda t}\left(1 - \mathrm{e}^{-\lambda t}\right)^{n-1}$, $\mathbb{E}(Y_t) = \mathrm{e}^{\lambda t}$, $\mathrm{Var}(Y_t) = \mathrm{e}^{\lambda t}\left(\mathrm{e}^{\lambda t} - 1\right)$; from k founders, the sum of k independent one-founder processes (negative binomial).

2.5 M5: Linear birth–death process, and time-inhomogeneous rates

Specified in §2.2.2.4; §2.2.3 paragraph "Linear birth–death rates".

State space. $\{0, 1, 2, \ldots\}$; state 0 absorbing. Non-zero rates out of $n \ge 1$:

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n. \tag{2.7}$$

$\mu = 0$ recovers the Yule process (M4).

Anchors. From $X_0 = N$ the mean is $N\mathrm{e}^{(\lambda-\mu)t}$. Extinction is certain for one founder when $\lambda \le \mu$ and has probability μ/λ when $\lambda > \mu$; from N founders, $(\mu/\lambda)^N$. The embedded chain is M1 with $q = \lambda/(\lambda + \mu)$.

Time-inhomogeneous variant. Per-capita rates $\lambda(t)$, $\mu(t)$. The mean closes: $\mathbb{E}(X_t \mid X_0 = n_0) = n_0 \exp\left(\int_0^t (\lambda(r) - \mu(r))\,\mathrm{d}r\right)$. The backward extinction probability $u(s,T) = \mathbb{P}(X_T = 0 \mid X_s = 1)$ satisfies the backward Riccati equation

$$-\frac{\partial u}{\partial s} = \mu(s) - \big(\lambda(s) + \mu(s)\big)u + \lambda(s)u^2, \qquad u(T,T) = 0, \tag{2.8}$$

whereas the forward-endpoint probability is $u_{\mathrm{f}}(t) = B(t)/(1+B(t))$ with $B(t) = \int_0^t \mu(s)\exp\left(\int_0^s (\mu(r) - \lambda(r))\,\mathrm{d}r\right)$ c
the two endpoints must not be conflated.

2.6 M6: The birth–death–catastrophe (BDC) process

Specified in Definition in §2.2.2.5; restated in §4.2; canonical restatement and notation in §5.2.1–5.2.2. Conventions: §4.2.3–4.2.4. Variant: several founders, §5.9..

State space. $\{0, 1, 2, \ldots\} \cup \{H\}$: the load within a compartment while it remains intact, with H a distinct absorbing rupture state. Both 0 and H are absorbing.

Transitions. Each unit independently gives birth at rate λ, dies at rate μ, and triggers catastrophe (rupture) at rate δ. For $n \ge 1$ the non-zero transition rates are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \qquad q_{n,H} = \delta n, \tag{2.9}$$

and, in first-step form,

$$\begin{aligned}
\mathbb{P}(\Delta X_t = 1,\ \Delta W_t = 0 \mid X_t = n) &= \lambda n\,\Delta t + o(\Delta t) && \text{(birth)}\\
\mathbb{P}(\Delta X_t = -1,\ \Delta W_t = 0 \mid X_t = n) &= \mu n\,\Delta t + o(\Delta t) && \text{(death)}\\
\mathbb{P}(X_{t+\Delta t} = H,\ W_{t+\Delta t} = n \mid X_t = n) &= \delta n\,\Delta t + o(\Delta t) && \text{(catastrophe)}\\
\mathbb{P}(\Delta X_t = 0,\ \Delta W_t = 0 \mid X_t = n) &= 1 - (\lambda + \mu + \delta)n\,\Delta t + o(\Delta t) && \text{(no event).}
\end{aligned} \tag{2.10}$$

Default initial condition: $X_t = 1$ (working point $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ in the Chapter 4 figures).

Release counter. With $\tau = \inf\{t : X_t = H\}$ the catastrophe time,

$$W_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \ge \tau, \end{cases} \qquad \mathcal{K} := X_{\tau-} \tag{2.11}$$

is the released count and $\mathcal{K}$ the release size at rupture. The left limit is essential.

Two conventions for rupture (§4.2.3). Sending rupture to 0 instead of H is *killing*; sending it to H is *catastrophe* (the thesis's convention). Killing makes internal extinction and rupture indistinguishable; with H separate one may distinguish the internal-extinction probability D, the rupture probability $1 - I$, and the productive probability $I_{\mathrm{fix}} = I - D$.

Numerical-load convention (§4.2.4). Wherever X_t appears arithmetically, $H \mapsto 0$; $\mathcal{K}$ (release size) is unrelated to $K(t) = \mathbb{E}(X_t^2)$ (second moment).

Characteristic roots and fixation functions (§5.2). With $\eta = (\lambda + \mu + \delta)/(2\lambda)$,

$$a = \eta + \sqrt{\eta^2 - \tfrac{\mu}{\lambda}}, \qquad b = \eta - \sqrt{\eta^2 - \tfrac{\mu}{\lambda}}, \qquad b < 1 < a, \tag{2.12}$$

$A := a - 1$, $B := 1 - b$, $\theta := \lambda(a - b) = \sqrt{(\lambda + \mu + \delta)^2 - 4\lambda\mu}$, and $AB = \delta/\lambda$. The no-catastrophe and internal-extinction probabilities

$$I(t) = \mathbb{P}(\text{no catastrophe by } t)\,, \qquad D(t) = \mathbb{P}(\text{internal extinction by } t,\ \text{no catastrophe})\,, \qquad I_{\text{fix}}(t) = I(t) - D(t \tag{2.13}$$

both satisfy the Riccati equation $\frac{\mathrm{d}I}{\mathrm{d}t} = \lambda(I - a)(I - b) = \mu - (\lambda + \mu + \delta)I + \lambda I^2$, differently initialised ($I(0) = 1$, $D(0) = 0$), with closed forms $I = \frac{aB + bAw}{B + Aw}$, $D = \frac{ab(w-1)}{aw - b}$, $I_{\text{fix}} = \frac{(a-b)^2 w}{(B+Aw)(aw-b)}$ at $w = \mathrm{e}^{\theta t}$. b is the probability of clearing internally without ever rupturing. Anchors used later: $V_\infty = \mathbb{E}(W_t(\infty)) = \frac{a(1-b)}{a-1} = \frac{aB}{A}$, $\mathbb{E}(\mathcal{K} \mid \text{catastrophe}) = a/(a-1)$, and the mean productive lifetime $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log \frac{a}{a-1}$.

Several founders (§5.9). From founding multiplicity k, the fixation functions become $I(\alpha)^k - D(\alpha)^k$ with release kernel $g_k(\alpha) = \delta K_k(\alpha)$; families founded independently factor until the first catastrophe, which stops all lineages at once.

Provenance note. The classical treatments cited in §5.2.2 allow immigration and distributed catastrophe sizes; the thesis specialises to the constant-rate, unit-catastrophe case above. A state-dependent catastrophe is not a mean-field hazard: $\mathbb{P}(\text{no catastrophe before generation } n) = \mathbb{E}\Big((1-\chi)^{\mathcal{E}_n}\Big)$ over cumulative exposure $\mathcal{E}_n = \sum_{i<n} \widetilde{X}_i$, and Jensen's inequality forbids replacing $\mathcal{E}_n$ by its mean (§2.2.2.5, Remark).

2.7 M7: Seasonal two-population process

Specified in §2.2.3.1.

State space. $\{0, 1, \ldots\}^2$: counts X_t, Y_t of two interacting populations.

Transitions. Time-inhomogeneous generator

$$q_{(x,y),(x+1,y)}(t) = b\mathcal{S}^+(t)\,x, \qquad q_{(x,y),(x-1,y+1)}(t) = exy, \qquad q_{(x,y),(x,y-1)}(t) = dy, \tag{2.14}$$

with seasonal multiplier $\mathcal{S}^+(t) = \max\{0,\, 1 + A\sin(\omega t)\}$ (clipped because a rate cannot be negative). For $A > 1$ each period $2\pi/\omega$ contains a closed season of length $(2/\omega)\arccos(1/A)$. On the axis $y = 0$ the X population is a time-inhomogeneous Yule process; in the interior the first moments involve $\mathbb{E}(X_t Y_t)$ and do not close. The deterministic comparison system is $\dot{x} = b\mathcal{S}^+(t)x - exy$, $\dot{y} = exy - dy$. Exact paths are simulated by thinning against $\mathcal{S}^+(t) \le 1 + A$.

2.8 M8: Logistic-environment type-count process

Specified in §2.2.3.2; full distribution in §9.2.

Driving environment. Total abundance $N(t)$ follows the logistic ODE $\frac{\mathrm{d}N}{\mathrm{d}t} = \varrho N\left(1 - \frac{N}{K}\right)$ (Chapter 9 notation; Chapter 2 uses r, C), with solution $N(t) = K\mathrm{e}^{\varrho t}/(\xi + \mathrm{e}^{\varrho t})$, $\xi = K/N_0 - 1$.

State space and transitions. $S_t \in \{1, 2, \ldots\}$, $S_0 = 1$, no removal; a pure birth process with time-dependent per-capita rate

$$q_{s,s+1}(t) = \beta(t)\, s, \qquad \beta(t) = \sigma\big(K - N(t)\big) = \sigma K\, \frac{\xi}{\xi + \mathrm{e}^{\varrho t}}. \tag{2.15}$$

Exactly a Yule process under a deterministic time change; one-way coupling (the type count does not feed back on N).

Law. With $\mathcal{B}(t) = \int_0^t \beta$, the marginal law is geometric, $\mathbb{P}(S_t = n) = \mathrm{e}^{-\mathcal{B}(t)}(1 - \mathrm{e}^{-\mathcal{B}(t)})^{n-1}$ for $n \geq 1$ (§9.2.4); $\mathbb{E}(S_t) = \mathrm{e}^{\mathcal{B}(t)}$ converges to the finite limit $(K/N_0)^{\sigma K/\varrho}$ (Chapter 2: $(C/N_0)^{\sigma C/r}$), and $\mathcal{B}(\infty) = \frac{\sigma K}{\varrho} \ln \frac{K}{N_0}$, so S_∞ is a proper geometric limit.

Related numerical variant (§2.2.3.2). A logistic-environment birth–death comparison with origination $\beta_{\mathrm{bd}}(t) = \sigma N'(t)$ and constant per-type removal ε is used for figure-level comparison only.

2.9 M9: Accumulation with gated release (coupled ODE–CTMC, two-way)

Specified in §2.2.4.1.

State space. $(A_t, R_t, G_t) \in [0, \infty) \times [0, \infty) \times \{0, 1\}$: accumulated amount, transferred amount, gate. A piecewise-deterministic Markov process.

Flow (between gate jumps).

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \rho - G_t\, cA_t, \qquad \frac{\mathrm{d}R_t}{\mathrm{d}t} = G_t\, cA_t, \tag{2.16}$$

with external input $\rho > 0$ and transfer rate $c > 0$.

Jumps. The gate opens at a state-dependent rate and closes at a constant one:

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0, A_t = a) = \eta a\, \Delta t + o(\Delta t), \qquad \mathbb{P}(G_{t+\Delta t} = 0 \mid G_t = 1) = \gamma\, \Delta t + o(\Delta t). \tag{2.17}$$

Pathwise balance: $A_t + R_t = A_0 + R_0 + \rho t$. The moment $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(A_t) = \rho - c\mathbb{E}(G_t A_t)$ is not closed. Illustration parameters: $\rho = 2$, $c = 4$, $\eta = 0.015$, $\gamma = 1.45$, $(A_0, R_0, G_0) = (5, 0, 0)$.

2.10 M10: Accumulation with surging gated release (mode-driven)

Specified in §2.2.4.2.

Specification. The flow and closing rate of M9 are retained; opening is autonomous,

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0) = \nu\, \Delta t + o(\Delta t), \tag{2.18}$$

so G_t is an autonomous two-state CTMC. The gate mean closes: $\mathbb{E}(G_t) \to \nu/(\nu + \gamma)$; the balance law is unchanged. Illustration parameters: as M9 with $\nu = 0.3$.

Context. §2.2.4 also states the three-way taxonomy of couplings (chain switching the vector field; prescribed trajectory driving the rates; two-way) with joint generator $(\mathcal{L}f)(\mathbf{y}, i) = F(\mathbf{y}, i) \cdot \nabla_{\mathbf{y}} f + \sum_{j \neq i} q_{ij}(\mathbf{y})[f(\mathbf{y}, j) - f(\mathbf{y}, i)]$; M9 is the two-way case and M10 the first case.

2.11 M11: Absorption-only model

Specified in Appendix 2.B, §2.B.1.

Setting. One compartment drawing particles from a surrounding medium; X_t particles inside, Y_t outside; $X_0 = 0$, $Y_0 = N$.

Transitions.

$$\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) = j\alpha\,\Delta t \qquad \text{(absorption)} \tag{2.19}$$

and no event otherwise. Particles are neither created nor destroyed ($p_{i,j}(t) = 0$ unless $i + j = N$).

Law. $X_t \sim \text{Binomial}(N,\ 1 - \mathrm{e}^{-\alpha t})$; single-particle PGF $G_1 = (1 - \mathrm{e}^{-\alpha t})x + \mathrm{e}^{-\alpha t}y$.

2.12 M12: Absorption–death model

Specified in §2.3.3.2 worked example.

Transitions. To the absorption channel of M11 add interior death:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}
\end{aligned}$$

with $X_0 = 0$, $Y_0 = N$. Single-particle state probabilities: $p_{0,1}(t) = \mathrm{e}^{-\alpha t}$, $p_{1,0}(t) = \frac{\alpha}{\mu-\alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})$, $p_{0,0} = 1 - p_{0,1} - p_{1,0}$; the N-particle PGF is the Nth power of the single-particle one. Generator-level PDE: $\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}$.

2.13 M13: Absorption–birth–death model

Specified in Appendix 2.B, §2.B.2–2.B.3; closed form in §2.B.2 Proposition.

Transitions. Absorption as M11/M12 plus interior birth:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = 1,\ \Delta Y = 0 \mid X_t = i) &= i\lambda\,\Delta t, && \text{(birth)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}
\end{aligned}$$

with $X_0 = 0$, $Y_0 = N$.

Closed form. With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\omega = \alpha/b_1$, $W = (x - x_+)/(x - x_-)$ and $\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda}\sum_{n\geq 1}\frac{\zeta^n}{n+\omega} = x_+ + (x_+ - x_-)\left({}_2F_1(1,\omega;\omega+1;\zeta) - 1\right)$,

$$G(x,y,t) = \left\{\mathrm{e}^{-\alpha t}\left[y - \Psi(W)\right] + \Psi\left(W\mathrm{e}^{b_1 t}\right)\right\}^N. \tag{2.20}$$

The parameter-regular convolution representation is $g(x,y,t) = \mathrm{e}^{-\alpha t}y + \alpha\mathrm{e}^{-\alpha t}\int_0^t \mathrm{e}^{\alpha v}F(x,v)\,\mathrm{d}v$ with F the interior birth–death PGF, and $G = g^N$. Resonance at $\alpha = n(\mu - \lambda)$, $n = 1, 2, \ldots$; the generating function stays regular there.

2.14 M14: Chained immediate transfer ($\mu = 0$ birth–catastrophe chain)

Specified in §5.10; described as "successive birth–death–catastrophe" in §8.2.1.

Specification. M compartments, all empty but one holding a single particle. Each runs the $\mu = 0$ BDC process (M6 with $\mu = 0$, so only births λn and catastrophe δn); on rupture the entire released load passes immediately and wholesale into a fresh compartment, founding the next process with that load as initial census. No extracellular phase, no clearance, no re-founding; all M ruptures occur almost surely.

Anchors. With $s = \delta/(\lambda+\delta)$ and $\rho = \lambda/(\lambda+\delta) = 1-s$: event types ignore the load (birth with probability ρ, catastrophe with probability s), the births before catastrophe in a compartment founded with any load are $G \sim \mathrm{Geom}_0(s)$, and the kth release size is $r_k = 1 + G_1 + \cdots + G_k$ with

$$\mathbb{P}(r_k = n) = \binom{n+k-2}{k-1} s^k \rho^{n-1}, \qquad n \geq 1. \tag{2.21}$$

The case $\mu > 0$ is recorded as open.

2.15 M15: Budding comparator compartment

Specified in §5.11; restated §6.2.2.

Specification. A budding compartment experiences two independent Poisson event streams: release of one particle at rate p and removal of the compartment at rate $d_{\mathcal{I}}$, continuing until removal.

Anchors.

$$S_{\mathrm{bud}}(\alpha) = \mathrm{e}^{-d_{\mathcal{I}}\alpha}, \qquad V_{\mathrm{bud}}(\alpha) = \frac{p}{d_{\mathcal{I}}}\left(1 - \mathrm{e}^{-d_{\mathcal{I}}\alpha}\right), \tag{2.22}$$

and the total number released is geometric on $\{0, 1, 2, \ldots\}$:

$$\mathbb{P}(\mathcal{K}_{\mathrm{bud}} = k) = \frac{d_{\mathcal{I}}}{d_{\mathcal{I}} + p}\left(\frac{p}{d_{\mathcal{I}} + p}\right)^k, \qquad k = 0, 1, 2, \ldots. \tag{2.23}$$

Structural contrast with rupture-based release (M6): here release and removal are independent events; in M6 they are the same event.

Chapter 3

The population layer (thesis Chapter 6)

3.1 M16: Standard two-variable population model (deterministic projection)

Specified in §6.2.1; three-variable version in §1.5.

Specification (fixed target count). Active compartments $\mathcal{I}$ and free particles $\mathcal{V}$ obey

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = p\,\mathcal{I} - c\,\mathcal{V}, \qquad R_0 = \frac{\gamma T\,p}{c\,d_{\mathcal{I}}}, \tag{3.1}$$

with target sites T fixed, founding rate γ, particle clearance c, release rate p, removal rate $d_{\mathcal{I}}$. *Deterministic*; the thesis shows it is the exact one-compartment limit of M15 and a projection of the renewal system M17, not a Markov model itself. §1.5 gives the full three-variable version $\frac{\mathrm{d}T}{\mathrm{d}t} = s - \gamma TV$, $\frac{\mathrm{d}I}{\mathrm{d}t} = \gamma TV - d_I I$, $\frac{\mathrm{d}V}{\mathrm{d}t} = pI - cV$.

3.2 M17: Kernel-driven renewal system

Specified in Kernels §6.3; system and exactness §6.4.1–6.4.2; skeleton §6.4.3.

Intracompartment layer. The load inside a compartment of age α follows M6. Two kernels:

$$S(\alpha) := I_{\mathrm{fix}}(\alpha) = \frac{(a-b)^2\,\mathrm{e}^{\theta\alpha}}{(B + A\mathrm{e}^{\theta\alpha})\,(a\mathrm{e}^{\theta\alpha} - b)}, \qquad g(\alpha) := \delta K(\alpha) = V'(\alpha), \tag{3.2}$$

the survival kernel (probability still productive at age α) and the release kernel (expected release flux at age α); g is explicit because a rupture fires at rate δX_α and releases X_α, giving $\delta\mathbb{E}[X_\alpha^2]\,\mathrm{d}\alpha$.

Population system. With incidence $i(t) = \gamma T\mathcal{V}(t)$ and $\mathcal{I}_0$ initially active compartments of age zero:

$$\mathcal{I}(t) = \mathcal{I}_0\,I_{\mathrm{fix}}(t) + \int_0^t i(t-\alpha)\,I_{\mathrm{fix}}(\alpha)\,\mathrm{d}\alpha = \mathcal{I}_0\,I_{\mathrm{fix}} + (i * I_{\mathrm{fix}})(t), \tag{3.3}$$

$$\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t}(t) = \mathcal{I}_0\,g(t) + \int_0^t i(t-\alpha)\,g(\alpha)\,\mathrm{d}\alpha - c\,\mathcal{V}(t) = \mathcal{I}_0\,g + (i * g)(t) - c\,\mathcal{V}(t). \tag{3.4}$$

These are the first moments *exactly* of the stochastic process in which each founding event starts an independent BDC compartment and each particle founds at rate γT or clears at rate c (linear incidence is what closes the hierarchy).

Skeleton. Any kernel pair (S, g) induces

$$\mathcal{I}(t) = \mathcal{I}_0\, S(t) + (i * S)(t), \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \mathcal{I}_0\, g(t) + (i * g)(t) - c\,\mathcal{V}(t), \tag{3.5}$$

with effective parameters $p_{\mathrm{eff}}(r) = \widetilde{g}(r)/\widetilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c) \int_0^\infty g$. Standing assumptions (A1)–(A5): constant T; continuum convention (particles not consumed on founding); synchronous zero-age founding cohort at $t = 0$; independent compartments, single founding; no spatial structure.

3.3 M18: Branching approximation of the early phase; the flooding parameter

Specified in §6.6.1–6.6.2.

Specification. Each released particle independently founds a new compartment at rate γT or is cleared at rate c, hence founds with probability

$$q = \frac{\gamma T}{\gamma T + c}. \tag{3.6}$$

The offspring of one active compartment is the release size $\mathcal{K}$ (M6) thinned by q: offspring PGF

$$G_{\mathrm{off}}(z) = b + \frac{\delta}{\lambda}\,\frac{y}{a - y}, \qquad y = 1 - q + qz, \tag{3.7}$$

including the mass b of compartments that never rupture. Mean $m = qV_\infty$ ($\approx R_0$ when $\gamma T \ll c$). The comparator (M15) is matched at equal total output $p/d_{\mathcal{I}} = V_\infty$; its thinned offspring law is geometric with the same mean.

Flooding parameter and criterion.

$$L := a(1 - b) = a - \frac{\mu}{\lambda}, \qquad \text{so that} \qquad V_\infty = \frac{L}{a - 1}. \tag{3.8}$$

For $m > 1$ the establishment extinction probabilities satisfy $z_{\mathrm{ext}}^{\mathrm{rupture}} - z_{\mathrm{ext}}^{\mathrm{bud}} = (L - 1)\left(\frac{1}{m} - 1\right)$, so one-shot release is advantageous if and only if

$$\delta(\lambda + \mu) > \lambda\mu \qquad \Longleftrightarrow \qquad \frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu} \tag{3.9}$$

(with the condition holding for all $\delta > 0$ when $\mu = 0$).

3.4 M19: Reset process (recurrent release)

Specified in §6.7.1.

Specification. While the compartment is alive, the load undergoes birth at rate λn, death at rate μn, optional immigration at rate ν, and a *dump that resets*: at rate δn the whole pool is released and the load returns to zero, after which it may re-accumulate; the compartment is removed at a separate, load-independent rate $d_{\mathcal{I}}$. Dumps do not absorb.

Kernels and anchors.

$$S(\alpha) = \mathrm{e}^{-d_{\mathcal{I}}\alpha}, \qquad g(\alpha) = \delta\,\mathrm{e}^{-d_{\mathcal{I}}\alpha}\,\mathbb{E}\Big[X_\alpha^2\Big], \qquad p_{\mathrm{eff}}(r) = (r + d_{\mathcal{I}})\,\widetilde{g}(r), \qquad d_{\mathcal{I},\mathrm{eff}} \equiv d_{\mathcal{I}}, \quad (3.10)$$

with $\mathbb{E}[X_\alpha^2]$ computed for the non-absorbing reset process. $R_0 = (\gamma T/c)\int_0^\infty g$ is finite only for $d_{\mathcal{I}} > 0$; in the mature-compartment limit, $p_\infty = \delta\,\mathbb{E}_\pi[X^2]$ for stationary law π.

3.5 M20: Partial release (boolean thinning)

Specified in §6.7.2.

Specification. Release events fire on the load-dependent clock δX and remove each unit independently with probability φ (the default); at mean-field level, with Q the total intracompartmental stock across productive compartments,

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \gamma T\,\mathcal{V} - d_{\mathcal{I}}\mathcal{I}, \qquad \frac{\mathrm{d}Q}{\mathrm{d}t} = \nu\mathcal{I} - \delta\varphi\,\frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}}Q, \qquad \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \delta\varphi\,\frac{Q^2}{\mathcal{I}} - c\,\mathcal{V}. \qquad (3.11)$$

With a load-independent clock $\lambda(X) = \delta$ instead, mean export is $\delta\varphi Q$ and the equations collapse to stock-and-export dynamics with $\varepsilon_{\mathrm{eff}} = \delta\varphi$: partial release is invisible in the mean field exactly when the clock is load-independent; in single-compartment release-count distributions it is always visible.

3.6 M21: Budding-kernel extension with sigmoid incidence

Specified in §6.7.3.

Specification. A non-catastrophe application: the mean load follows an immigration-like $\bar{x}' \approx \nu - \mu\bar{x}$. The skeleton of M17 carries the application on a different kernel pair $(S_{\mathrm{b}}, g_{\mathrm{b}})$ — probability of still being in a productive lineage, and density of productive sojourn times — with a multi-stage delay (near-zero release for small α, then rising and falling) supplying the eclipse; the effective-parameter map carries over verbatim. Near the observed establishment threshold the data favour a sigmoid incidence:

$$f(\mathcal{V}, T) = \gamma T\,\mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \qquad (3.12)$$

under which the first-moment exactness of M17 no longer holds.

Chapter 4

Two-type extension (thesis Chapter 7)

4.1 M22: Two-type birth–death–conversion with type-specific catastrophe rates

Specified in §7.2.1 (definition and rate list), §7.2.2 (occupation-time representation), §7.2.4 (dimensionless form).

State space. $(x, y) \in \mathbb{Z}^2_{\geq 0}$ before absorption; a single global absorbing state (R, R).

Transitions. Type 1 gives birth at rate λ_1, dies at rate μ_1, and converts irreversibly to type 2 at rate ν; type 2 gives birth at rate λ_2 and dies at rate μ_2; all rates non-negative. Before absorption, the non-zero transition rates from (x, y) are

$$\begin{array}{lll}
(x,y) \longrightarrow (x+1,y) & \text{at rate} & \lambda_1 x, \\
(x,y) \longrightarrow (x-1,y) & \text{at rate} & \mu_1 x, \\
(x,y) \longrightarrow (x-1,y+1) & \text{at rate} & \nu x, \\
(x,y) \longrightarrow (x,y+1) & \text{at rate} & \lambda_2 y, \\
(x,y) \longrightarrow (x,y-1) & \text{at rate} & \mu_2 y, \\
(x,y) \longrightarrow (\mathsf{R},\mathsf{R}) & \text{at rate} & \delta_1 x + \delta_2 y,
\end{array} \tag{4.1}$$

the last being the catastrophe: one unit contributes a type-specific rate, and the first trigger absorbs the whole process. One reading assigns the types to two internal states of a unit within a compartment, catastrophe representing irreversible loss of containment.

Object of study. $S(t) = \mathbb{P}_{(1,0)}\{\tau_c > t\}$ and $G(t) = \mathbb{P}_{(0,1)}\{\tau_c > t\}$: no-catastrophe probabilities, not population-survival probabilities. They admit the occupation-time representation

$$S(t) = \mathbb{E}_{(1,0)}\left[\exp\left\{-\int_0^t (\delta_1 X_s + \delta_2 Y_s)\,\mathrm{d}s\right\}\right], \qquad G(t) = \mathbb{E}_{(0,1)}\left[\exp\left\{-\int_0^t (\delta_1 X_s + \delta_2 Y_s)\,\mathrm{d}s\right\}\right], \tag{4.2}$$

and satisfy the triangular backward system

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \qquad S(0) = 1,\ S'(0) = -\delta_1, \tag{4.3}$$

$$\frac{\mathrm{d}G}{\mathrm{d}t} = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \qquad G(0) = 1,\ G'(0) = -\delta_2. \tag{4.4}$$

Dimensionless form: time in units of λ_1; rates $\hat{\mu}_i = \mu_i/\lambda_1$, $\hat{\nu} = \nu/\lambda_1$, $\hat{\delta}_i = \delta_i/\lambda_1$, $\hat{\lambda}_2 = \lambda_2/\lambda_1$. The main formula assumes $\hat{\lambda}_2 > 0$, $\hat{\delta}_2 > 0$, $\hat{\nu} > 0$; the zero-birth and zero-catastrophe boundaries are treated separately, and with the appropriate initial data the zero-catastrophe case recovers the two-type branching equations of Antal and Krapivsky. §7.5 compares four catastrophe-rate regimes (per-type constant, type-2 only, and mixtures) exactly.

Chapter 5

Depletion models (thesis Chapter 8)

5.1 M23: Replicator–suppressor process

Specified in §8.3.2 (definition); first-step analysis §8.3.3; fates §8.3.4.

Setting. A compartment contains a replicating population and a finite store of suppressor units; suppressors are consumed one-for-one as they act, and the store is never replenished.

State space. $\mathcal{S} = \{(i, j) : i \in \mathbb{N} \cup \{0\},\ j \in \{0, \dots, N\}\}$: X_t replicating units in the compartment, Q_t remaining suppressor units (store size N fixed at founding).

Transitions. For $(i, j) \in \mathcal{S}$:

$$\mathbb{P}(\Delta X_t = 1,\ \Delta Q_t = 0 \mid X_t = i,\ Q_t = j) = \lambda i\, \Delta t + o(\Delta t) \qquad \text{(birth)}, \tag{5.1}$$

$$\mathbb{P}(\Delta X_t = -1,\ \Delta Q_t = -1 \mid X_t = i,\ Q_t = j) = \varsigma ij\, \Delta t + o(\Delta t) \qquad \text{(suppression)}, \tag{5.2}$$

$$\mathbb{P}(\Delta X_t = 0,\ \Delta Q_t = 0 \mid X_t = i,\ Q_t = j) = 1 - (\lambda i + \varsigma ij)\, \Delta t + o(\Delta t) \qquad \text{(no event)}. \tag{5.3}$$

The product ςij is the only nonlinearity. The column $X_t = 0$ is extinction; the row $Q_t = 0$ is a spent store, from which the process is a pure birth process and survival is certain. Interest centres on $\mathbb{P}(\text{survival}) = 1 - \lim_{t\to\infty} \mathbb{P}(X_t = 0)$; survival is certain whenever the founding cohort outnumbers the store ($i > j$), and in the contested region $1 \le i \le j$ the survival map $\pi_{i,j}$ is computed by first-step analysis.

5.2 M24: Fixed-removal extension of the chained process

Specified in §8.4.1–8.4.2; assumptions in §8.4.1 Remarks.

Specification. The chained-rupture picture (M14's complete transfer, or dispersal to new compartments) is extended by one assumption: after each release event, a fixed count of units is removed. The removal rule is

$$\text{removal} = \begin{cases} r, & r \le \vartheta, \\ \vartheta, & r > \vartheta, \end{cases} \qquad \text{that is, } \min(r, \vartheta), \tag{5.4}$$

with r the release size and ϑ the fixed removal budget (an idealisation of a saturating casualty law $\vartheta(r) \sim \log(r + 1)$). Under the framing used for the calculation, each surviving released unit founds a new compartment.

Assumptions. Remover-agent prevalence resets between release events (steady state, uniform distribution — this makes ϑ the same at every delivery and the independence exact); the release-size distribution is that of M6 including internal-death outcomes ($\mathbb{E}(r) = V_\infty$).

Quantity of interest. The expected replication ratio

$$\mathcal{R}(\vartheta) = \mathbb{E}(\max(0,\, r - \vartheta)) = \big[\mathbb{E}(r \mid r \geq \vartheta) - \vartheta\big]\mathbb{P}(r \geq \vartheta)\,, \tag{5.5}$$

the expected number of newly founded compartments created by one founding event. Deliberately not the R_0 of M17–M18; the relation is stated in §8.5.1.

Chapter 6

Models with non-constant rates (thesis Chapters 9–10)

6.1 M25: Competitive-reversal chain (evolving resistance)

Specified in §9.3.2; critical point §9.3.4.

State space. $A_t \in \{0, 1, \dots, N\}$ in discrete time, two populations with $A_t + B_t = N$; $i = 0$ and $i = N$ absorbing. The triple $(A_t, R_{a,t}, R_{b,t})$ is Markov; A_t alone is not.

Transition kernel. For $0 < i < N$:

$$\mathbb{P}(A_{t+1} = j \mid A_t = i) = \begin{cases} q_a = \dfrac{i}{N}(1 - p_b), & j = i + 1, \\ q_b = \left(1 - \dfrac{i}{N}\right)(1 - p_a), & j = i - 1, \\ q_c = \dfrac{i}{N} p_b + \left(1 - \dfrac{i}{N}\right) p_a, & j = i, \\ 0 & \text{otherwise.} \end{cases} \tag{6.1}$$

Blocking capacities

$$p_a = \max\left\{0,\, 1 - \frac{R_b}{R_a}\right\}, \qquad p_b = \max\left\{0,\, 1 - \frac{R_a}{R_b}\right\}, \tag{6.2}$$

so at most one of p_a, p_b is non-zero. Resistances are driven up when the population is in a minority:

$$R_{k,t+1} = R_{k,t} + \varepsilon\, \Delta R_{k,t}, \qquad \Delta R_{a,t} = \max\left\{0, \left(\frac{N - A_t}{A_t}\right)^{\alpha}\right\}, \quad \Delta R_{b,t} = \max\left\{0, \left(\frac{A_t}{N - A_t}\right)^{\alpha}\right\}, \tag{6.3}$$

with $R_{a,0} = R_{b,0} = 1$ without loss (only ε / R_0 matters). The critical state $q_a = q_b$ is $i_c = N\frac{1-p_a}{2-p_a}$ ($p_a > 0$) or $N\frac{1}{2-p_b}$ ($p_b > 0$), and is unstable; it is a function of the path, not the parameters alone.

6.2 M26: Fitness-coupled type-origination variant

Specified in §9.3.6; recorded as specified but never simulated.

Specification. The kernel of M25 recast with fitness in place of resistance, a type count carried alongside and acting back on fitness:

$$\mathbb{P}(\Delta A_t = 1) = \frac{A_t}{N}\,\frac{f_a}{\bar f}, \qquad \mathbb{P}(\Delta A_t = -1) = \left(1 - \frac{A_t}{N}\right)\frac{f_b}{\bar f}, \tag{6.4}$$

with $\bar f$ the mean fitness across the domain; fitness updates

$$f_a(t+\Delta t) = f_a(t) + \varepsilon\left(\frac{B_t}{A_t}\right)^{\alpha} - \gamma S_t^{(a)}, \qquad f_b(t+\Delta t) = f_b(t) + \varepsilon\left(\frac{A_t}{B_t}\right)^{\alpha} - \gamma S_t^{(b)}, \tag{6.5}$$

and per-population type counts evolving as birth–death processes with constant removal μ_a and dominance-and-fitness origination

$$\mathbb{P}\Big(\Delta S_t^{(a)} = 1\Big) = \beta_a S_t^{(a)}\,\Delta t, \qquad \beta_a = \left(\frac{A_t}{N}\right)^{\zeta} f_a. \tag{6.6}$$

The penalty $-\gamma S$ is the endogenous weakening of the leader; ζ controls how firmly dominance must be established before variation is released.

6.3 M27: Rise–run–ruin–rejuvenation (RRRR)

Specified in §9.4.2.

State space. $X_t \in \{0, 1, 2, \ldots\}$ replicating agencies with an auxiliary real potential $V(t)$; the pair $(X_t, V(t))$ is Markov, X_t alone is not. Described in the source as "a birth–death–catastrophe process with catastrophe rate $\delta X_t / V(t)$ and an auxiliary ODE for V".

Specification. Potential flows in and is consumed:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t. \tag{6.7}$$

Given V, the count moves by

$$\begin{aligned}
\mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda X_t\,\Delta t + o(\Delta t) && \text{(proliferation)},\\
\mathbb{P}(X_{t+\Delta t} = 0 \mid X_t) &= \delta\,\frac{X_t}{V(t)}\,\Delta t + o(\Delta t) && \text{(catastrophe)},\\
\mathbb{P}(X_{t+\Delta t} = 1 \mid X_t = 0) &= \omega\,\Delta t + o(\Delta t) && \text{(revival)},\\
\mathbb{P}(X_{t+\Delta t} = X_t) &= 1 - \left(\lambda X_t + \delta\frac{X_t}{V(t)} + \omega\mathbf{1}\{X_t = 0\}\right)\Delta t + o(\Delta t) && \text{(no event)}.
\end{aligned} \tag{6.8}$$

Catastrophe is driven by per-capita scarcity; revival keeps 0 from absorbing, making the process recurrent. V stays positive for all time (the hazard diverges as $V \downarrow 0$). Illustration parameters: $\phi = 4$, $c = 0.8$, $\lambda = 0.5$, $\delta = 0.25$, $\omega = 0.15$.

6.4 M28: Multiplicative-dissipation automaton

Specified in §9.5 (state, competition, mutation and potential, fragmentation).

State space. An $L \times L$ lattice ($L = 256$; Moore neighbourhood, periodic boundary, torus), each cell empty or occupied. Every occupied cell carries a type and, within it, a sub-type. For each extant type i: accumulated potential V_i (real), occupied cells N_i, extant sub-types $X_i \le N_i$. Continuous updates (potential, mutation) advance in chunks between discrete contests (one attempt per site per sweep).

Competition. A cell ζ carrying sub-type u of type i has strength

$$w(\zeta) = V_i \times (n_\zeta + 1)^\theta \times \mathrm{e}^{-\nu|u|/\kappa_i}, \qquad \kappa_i = \frac{N_i}{X_i}, \tag{6.9}$$

with n_ζ matching neighbours at the contest level and $|u|$ the area of u; the contest goes to ζ with probability $w(\zeta)/(w(\zeta)+w(\zeta'))$. Between sub-types of one type the potential cancels; between distinct types the rarity factor is omitted, $\mathbb{P}(A \text{ wins}) = \frac{V_A(n_A+1)^\theta}{V_A(n_A+1)^\theta + V_B(n_B+1)^\theta}$. Contests against empty ground: fixed strength d, an occupied cell takes a contested empty neighbour with probability $1-d$, empty ground never invades.

Mutation and potential. Sub-types arise within a type at type-wide rate $R_i^{\mathrm{mut}} = m\,V_i N_i$ (per-cell rate mV_i). Each occupied cell supplies potential at rate φ and each extant sub-type costs c:

$$\frac{\mathrm{d}V_i}{\mathrm{d}t} = \varphi N_i - c\,X_i, \qquad \text{floored at } 0, \tag{6.10}$$

so potential can fall at all only if $c > \varphi$.

Fragmentation. When a type carries too many sub-types it fragments: every sub-type becomes an independent type at once (nothing dies; the interaction law changes). Two triggers, tested in order: deterministically if the potential update would take V_i to zero or below; otherwise with hazard

$$R_i^{\mathrm{frag}} = \delta\,\frac{X_i}{V_i}, \tag{6.11}$$

M27's catastrophe rate with the count of individuals replaced by the count of types. No revival term is needed: the catastrophe restarts the process.

6.5 M29: Birth–conversion model, and the vitality-coupled variant

Specified in Appendix 9.A; simulated but not analysed.

State space. X_t unconverted individuals, Y_t converted.

Transitions.

$$\mathbb{P}(\Delta X_t = 1,\ \Delta Y_t = 0 \mid X_t, Y_t) = \lambda X_t\,\Delta t + o(\Delta t) \qquad \text{(birth)}, \tag{6.12}$$

$$\mathbb{P}(\Delta X_t = -1,\ \Delta Y_t = 1 \mid X_t, Y_t) = (\delta + \chi Y_t) X_t\,\Delta t + o(\Delta t) \qquad \text{(conversion)}, \tag{6.13}$$

$$\mathbb{P}(\Delta X_t = 0,\ \Delta Y_t = -1 \mid X_t, Y_t) = \mu Y_t\,\Delta t + o(\Delta t) \qquad \text{(clearance)}. \tag{6.14}$$

δ is the spontaneous-conversion rate (linear in X_t) and χ the contact-conversion coefficient (bilinear X_tY_t); the two mechanisms superpose in one channel. Illustration parameters: $(\lambda, \delta, \chi, \mu) = (1,\ 0.1,\ 0.02,\ 10)$.

Vitality-coupled variant. Carry $V(t)$ with $\frac{\mathrm{d}V}{\mathrm{d}t} = \phi - c\,X_t$ and replace the middle channel by

$$\mathbb{P}(\Delta X_t = -1,\ \Delta Y_t = 1 \mid X_t, Y_t) = \left(\frac{\delta}{V(t)} + \chi Y_t\right) X_t\,\Delta t + o(\Delta t), \tag{6.15}$$

the other channels unchanged; a population that grows large drives V down and raises its own loss rate. Not simulated.

6.6 M30: Public-good accumulation process

Specified in Appendix 9.B.

State space. X_t units within one compartment and $P(t)$ the public-good concentration.

Specification.

$$\begin{aligned} \frac{\mathrm{d}P}{\mathrm{d}t} &= g\,X_t - c, \\ \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda(t)\,X_t\,\Delta t + o(\Delta t), \qquad \lambda(t) = \lambda_0 + \alpha P(t), \\ \mathbb{P}(\Delta X_t = -1 \mid X_t) &= \mu\,X_t\,\Delta t + o(\Delta t), \end{aligned} \tag{6.16}$$

with production rate g per unit, clearance c, base birth rate λ_0, public-good advantage α, death rate μ. The regime of interest is $\mu > \lambda_0$: subcritical alone, supercritical once enough good has accumulated.

Embedded Galton–Watson reading. In a compartment-to-compartment sequence the process is a Galton–Watson process, supercritical exactly when $\lim_{t\to\infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \frac{1}{2}$; the minimum founding cohort is

$$K^* = \inf\left\{K : \lim_{t\to\infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \tfrac{1}{2}\right\}. \tag{6.17}$$

Illustration parameters: $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$, so P falls whenever $X_t < c/g = 3$.

6.7 M31: Three-environment birth–death ordering

Specified in §10.3.1.

Specification. Replication in each of three environments indexed $i \in \{1, 2, 3\}$ in increasing order of hostility — treated as a simple birth–death process (M5) with rates (β_i, μ_i) — with

$$\mu_3 > \mu_2 > \mu_1 \qquad \text{and} \qquad \beta_1 = \beta_2 = \beta_3 = \beta, \tag{6.18}$$

so the net rates order as $\beta - \mu_1 > \beta - \mu_2 > \beta - \mu_3$. A consistency argument, not a derivation: the ordering is inferred from the strategy it is meant to explain. (The surrounding discussion notes that with a depletable suppressor store, as in M23, the μ_i are in reality declining functions of the cohort's history.)